Advanced stereochemistry

Chapter – 2 – Conformation and configuration
Cyclic compounds.

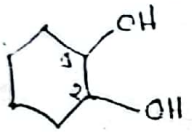
[2.1]

What is configuration ?

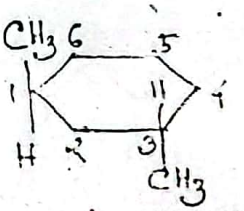
Configuration:
The arrangement of atoms or groups around the dissymmetric or asymetric carbon atom that characterises a particular steriaisomer is called configuration.

Example: Its isomers can separated easily.
Interconversion not possible.

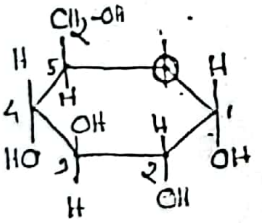
(1)



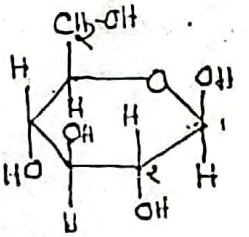
Cyclohexane diol – 1(R). 2(S)



1(S). 3(S) – Dimethylcyclohexane



α – D – Glucose



β – D – Glucose

Fig:- Anomers of glucose

2. What is conformation ?

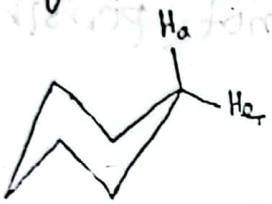
Conformation:-

Different arrangement of atoms (or groups) that can be converted into one another by rotation about single bonds are called conformation.

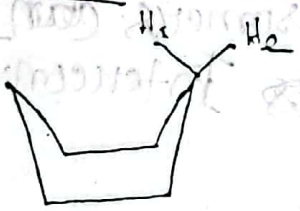
(i) Its isomers can not be separated easily.

Example: (ii) interconversion occur readily/easily.

Conformation of cyclohexane

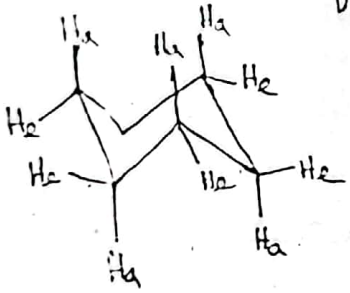


Chair conformation

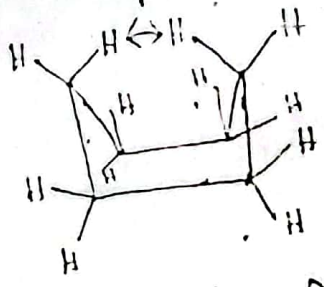


Boat conformation

3. Boat conformation is less stable (Higher energy) than Chair conformation. – Explain.



Chair conformation



Boat conformation

① Eclipsed bonds produce torsional strain (rise energy

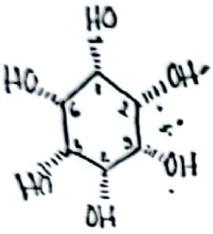
④ Steric strain of flagpole hydrogen

Draw the possible structure of inositols. [2007]

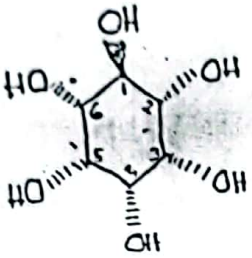
Inositols:

Inositols is a hexahydroxy substituted cyclohexane.

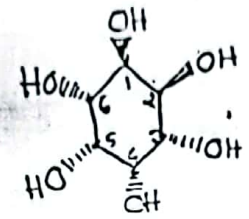
The possible structure of inositols:-



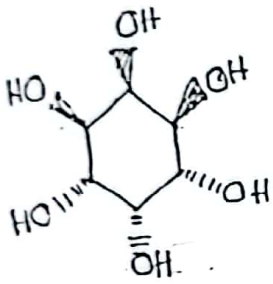
cis-



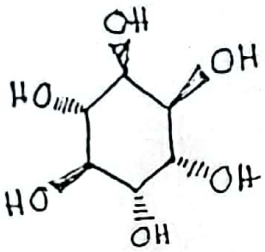
epi-



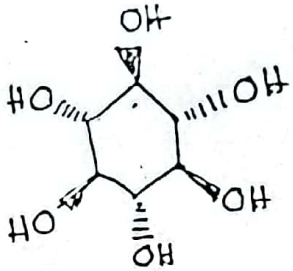
Allo-



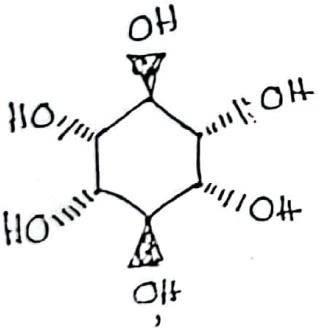
neo-



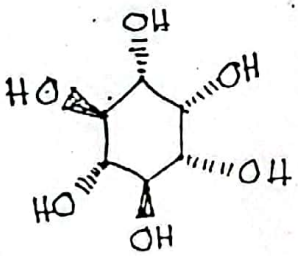
Chiro-



scyllo-



muco-



myo-

CEA NCSMM

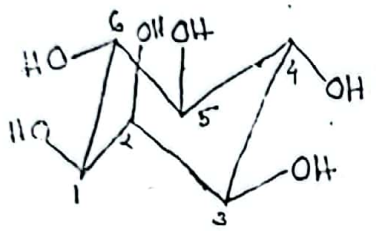
CeA 1dCf Chemistry Subjects More, More

MUKTA SAHA

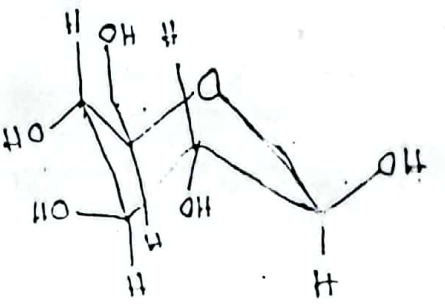
Pseudosugar:

Myoinositol form of cinnosetols as is called pseudosugar.

The structure of pseudosugar:



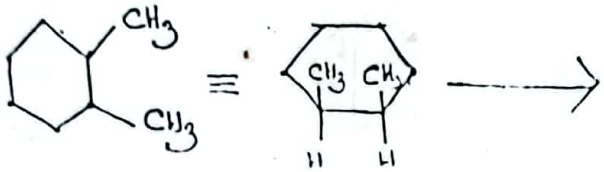
Pseudosugar.



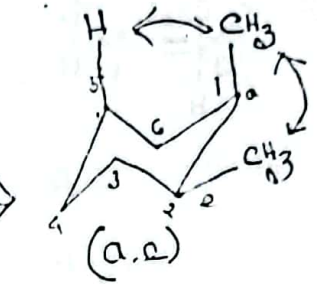
Sugar

⑮ Conformational analysis of disubstituted cyclohexane derivative.

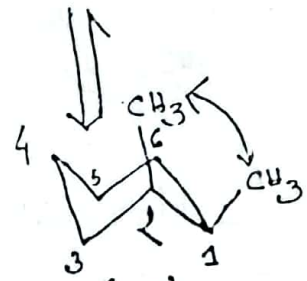
① 1,2-Dimethylcyclohexane

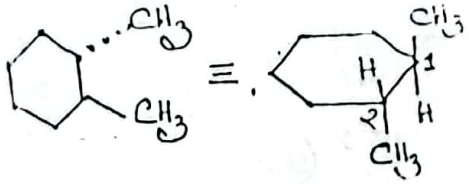


Cis-1,2-Dimethylcyclohexane

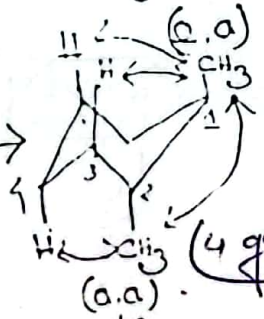


(a.e)





Trans-1,2-Dimethyl cyclohexane



(e.a)

(a.a) (4 gauch interaction)



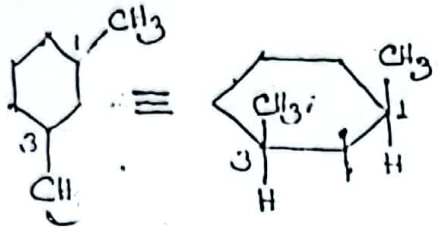
(e.e)

Stability:

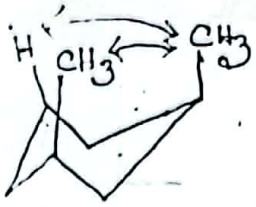
(a.a) < (a,e) < (e e)

MUKTA SAHA

1,3-Dimethylcyclohexane :-



cis-

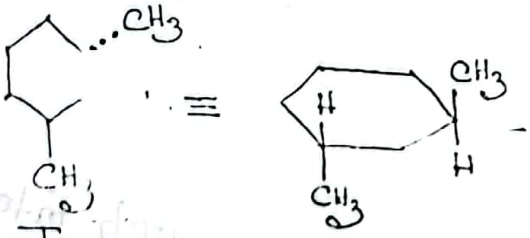


(a,a)

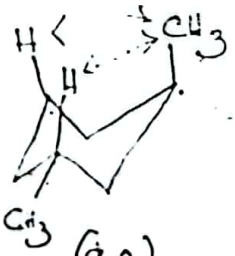


e,e

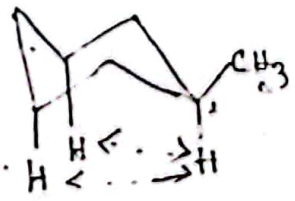
(Most stable)



Trans-



(a,e)

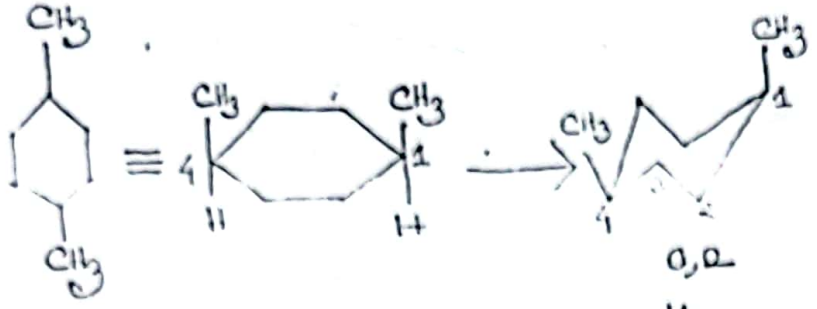


(e,a)

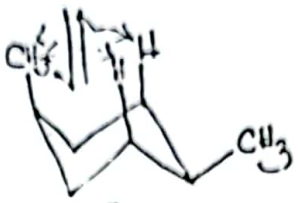
Stability :

Cis (a,a) < Trans (a,e) < Cis (e,e)

1,4-Dimethylcyclohexane:-

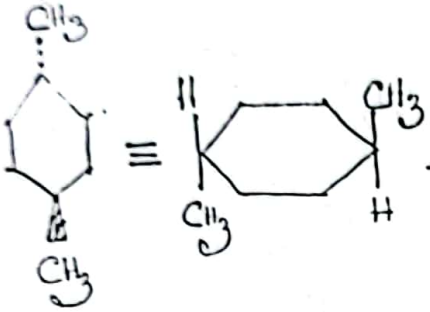


Cis-

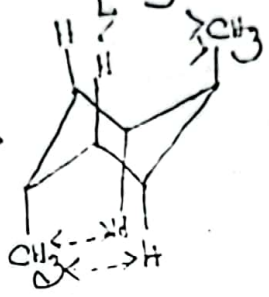


(e,a)

Equally stable



Trans



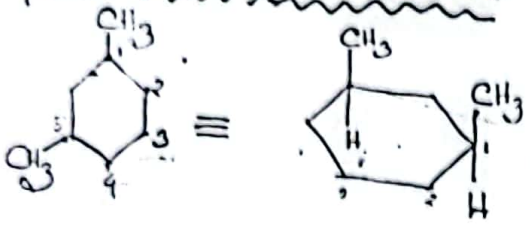
(a,a)



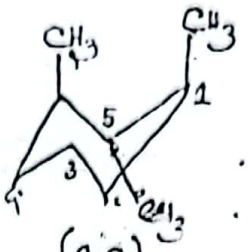
(e,e)

Most stable

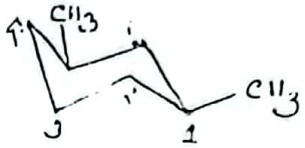
1,5-Dimethylcyclohexane:-



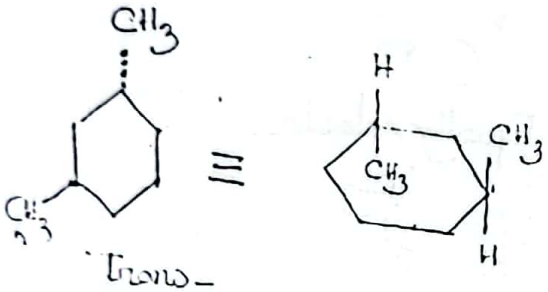
Cis.-



(a,a)



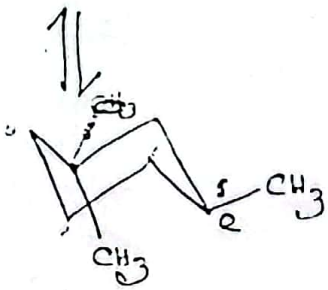
(e,e) (stable)



Trans-



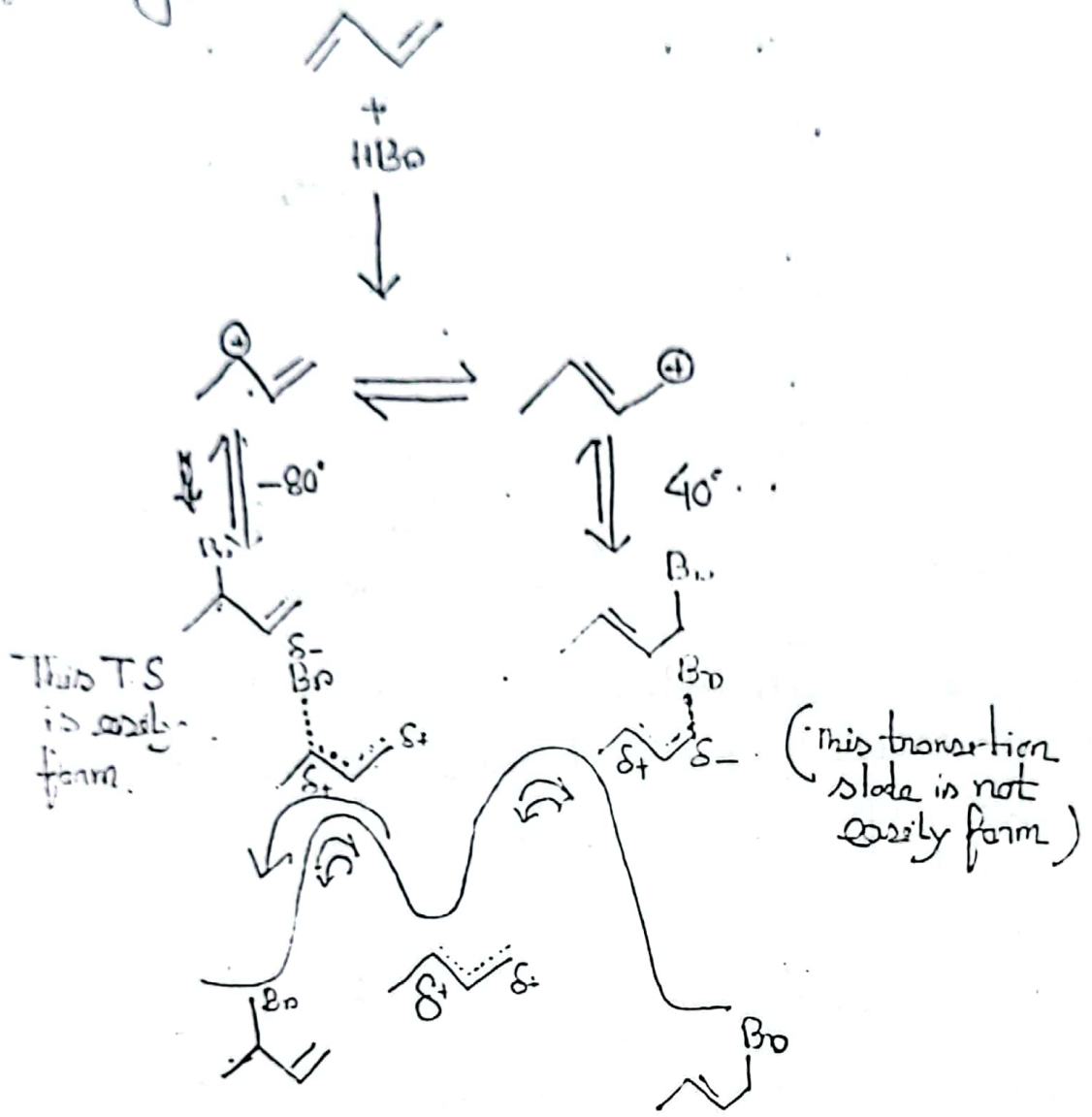
(a,e)



(e,a)

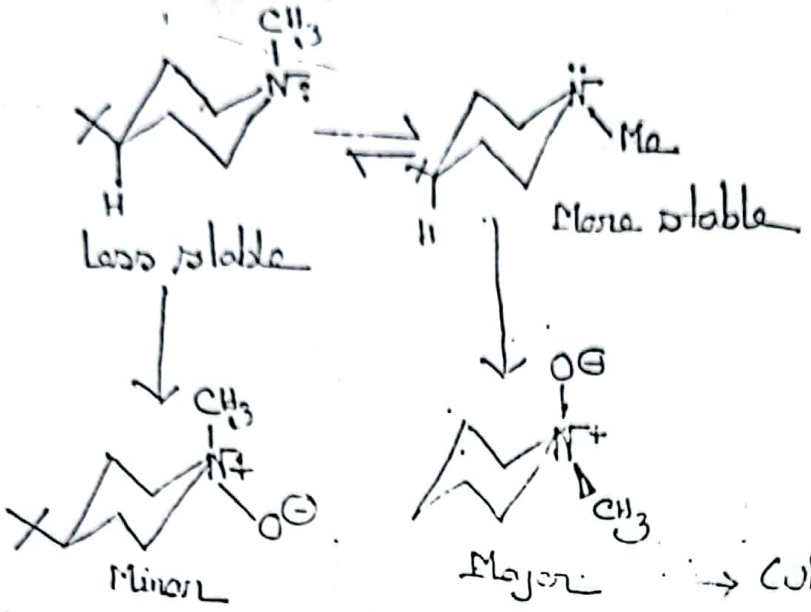
The Curtin Hammet principle:-

1 Chlor 1,3 - Butadine react with HBr to produce the following:

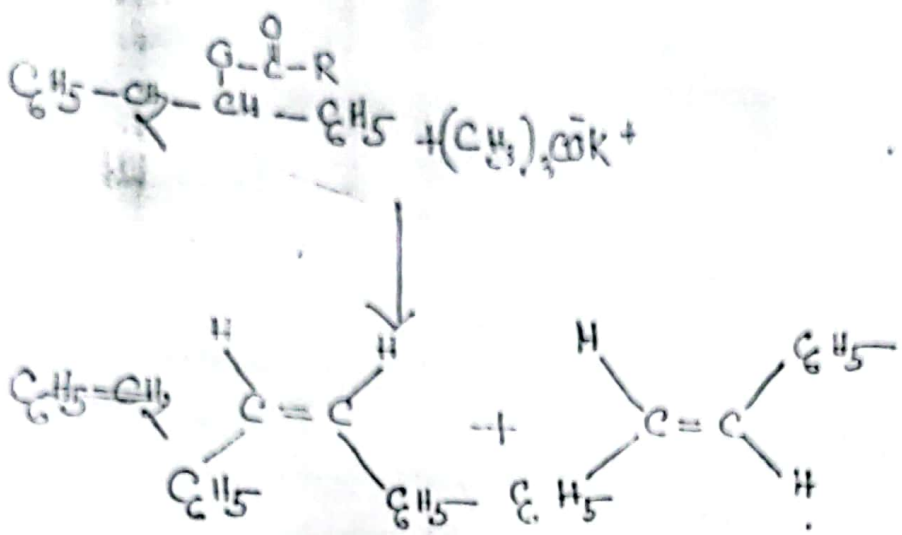


that can be
le bonds

2. Oxidation of N-piperidine :



Axil The axial-N-methyl compound are have high ly steric hindrance because of this the oxidation of axial N-methyl compound is slow Comparatively.

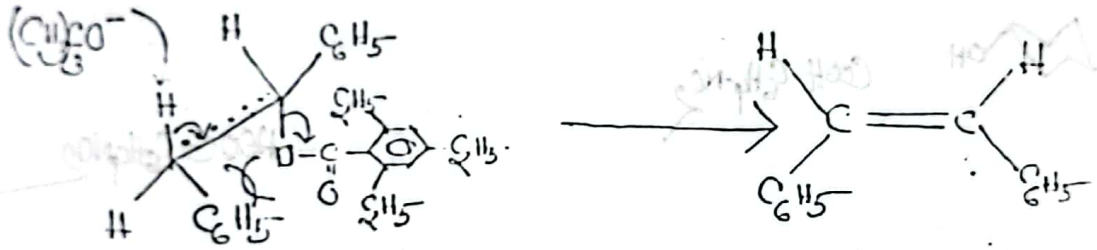
3. 

Cis. Stilbene Trans- Stilbene

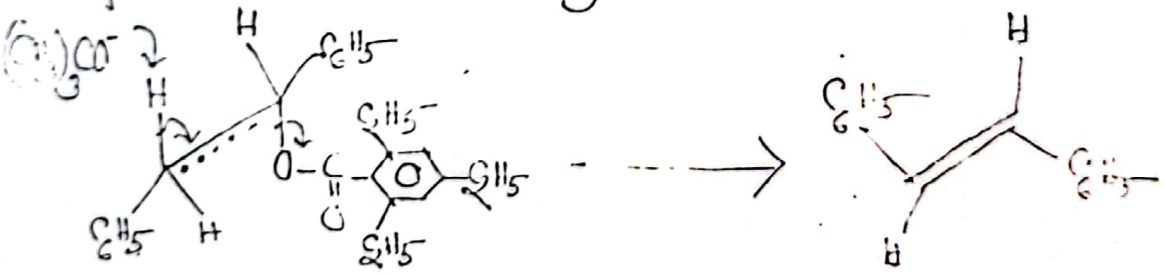
Here

$$\frac{\text{Trans- Stilbene}}{\text{Cis- Stilbene}} = 100$$

Explain - why

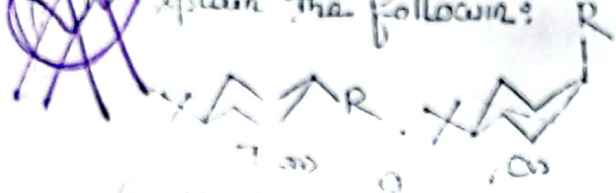


High energy needed for this transition state so Cis product does not easily obtained.



This transition state is easily formed

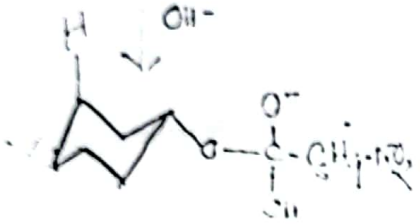
Explain the following:

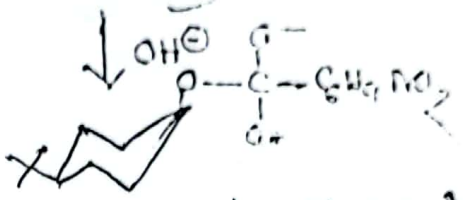


(i) If R = $-O-\overset{O}{\overset{\|}{C}}-C_6H_4NO_2$-p then $\frac{K_{trans}}{K_{cis}} = 2.5$

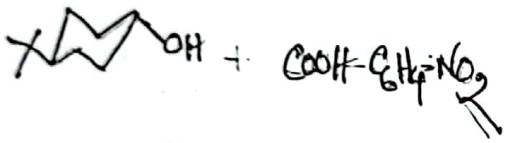




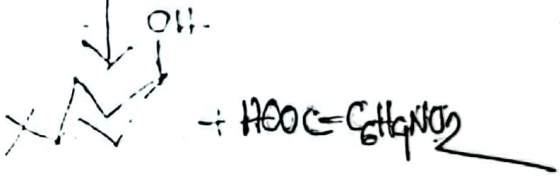




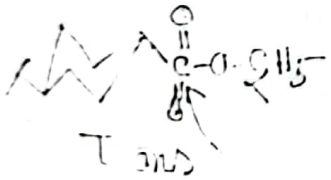
less steric hindrance

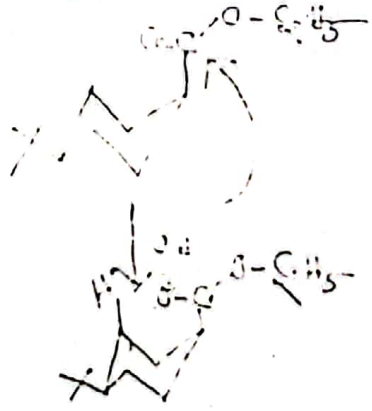


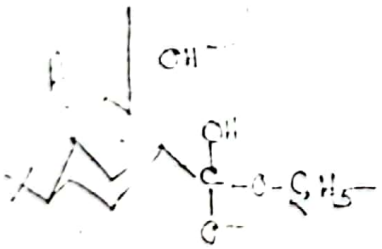
More steric hindrance



(ii) If R = $-\overset{O}{\overset{\|}{C}}-O-C_2H_5$ then $\frac{K_{trans}}{K_{cis}} = 10$.



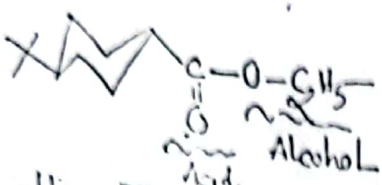




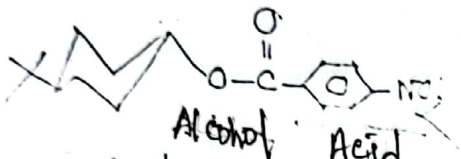
Less steric hindrance

More steric hindrance

(ii) $\frac{K_{trans}\left(-\overset{O}{\overset{\|}{C}}-O-C_2H_5\right)}{K_{trans}\left(-\overset{O}{\overset{\|}{C}}-C_6H_4NO_2\text{-}P\right)} = \frac{20}{2.5}$



Here Transition state ester is directly attached instead of alcohol, So $K_{trans}\left(-\overset{O}{\overset{\|}{C}}-O-C_2H_5\right)$ is 20 than



Here alcohol is directly attached with ring instead of ester

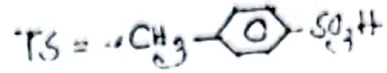
Explain the following :



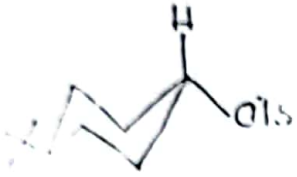
Trans



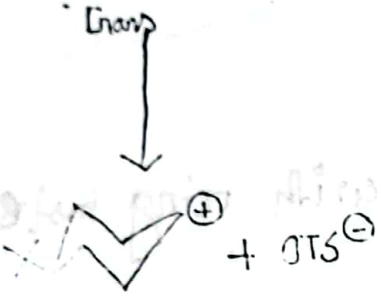
Cis

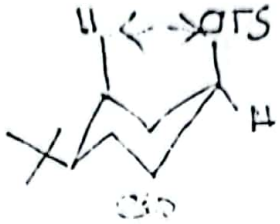
TS = 

(i) In S_N1 reaction $\frac{K_{cis}}{K_{trans}} = \frac{3or4}{1}$ -- Explain



Trans

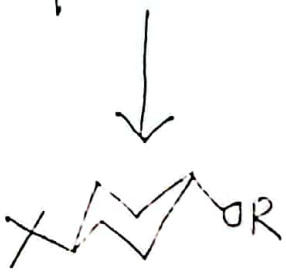




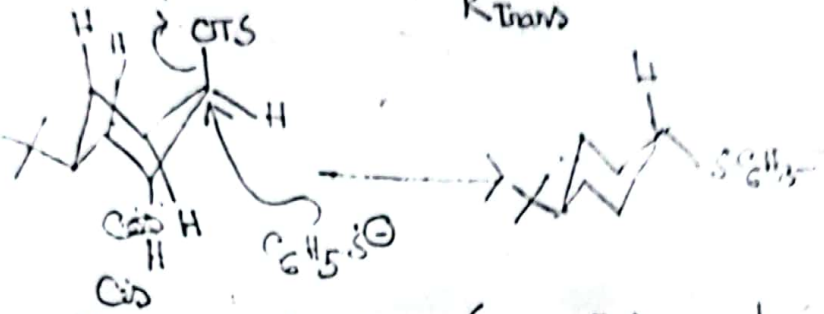
Cis



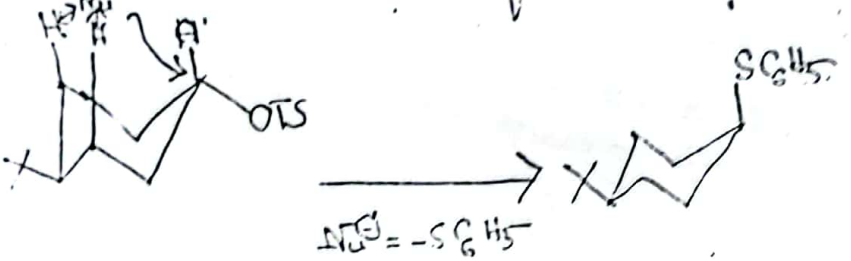
Easy easy form because of steric assistance



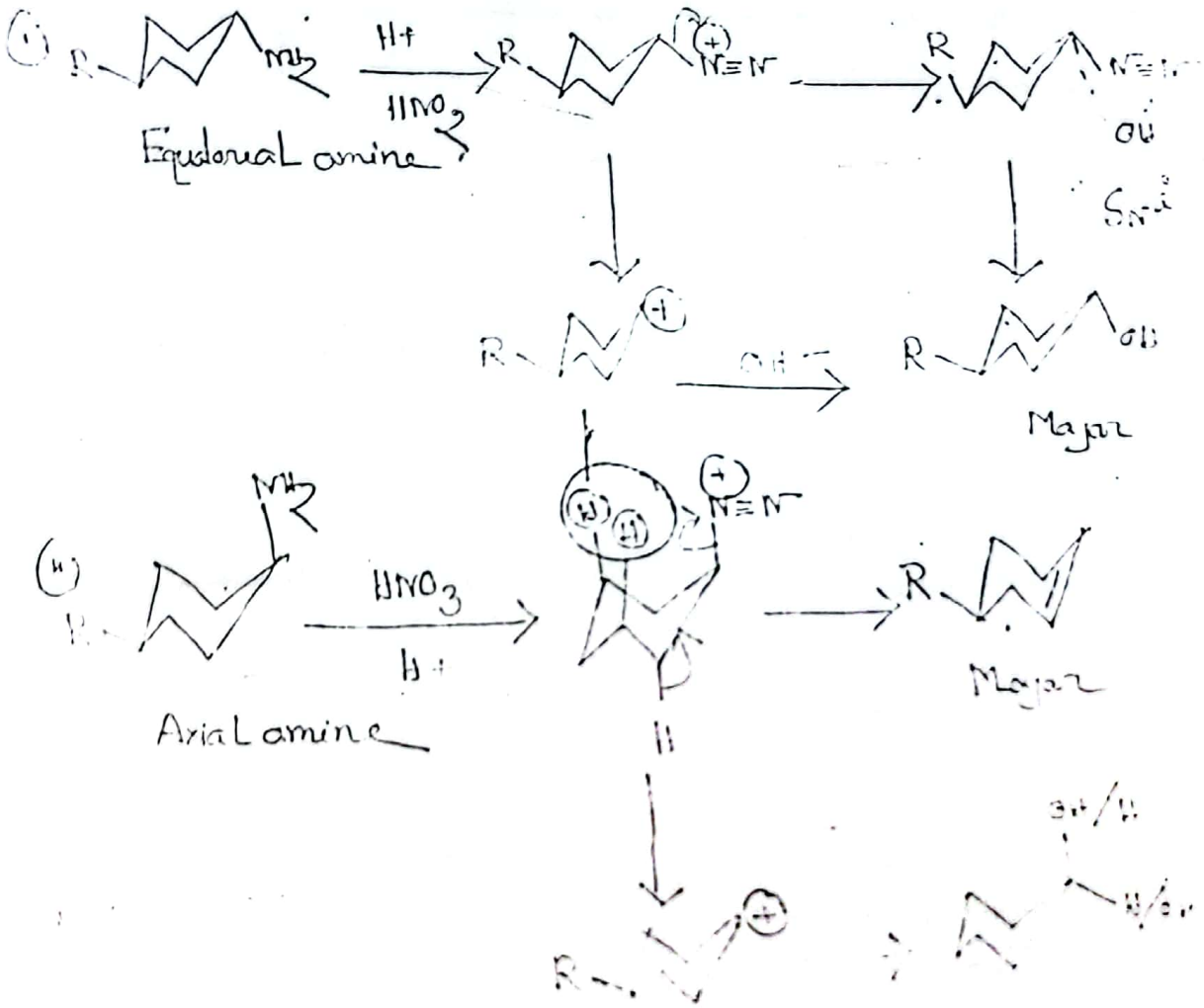
If S_N2 mode then $\frac{k_{cis}}{k_{trans}} = 31$ (b)



Here nucleophile ($C_6H_5S^{\ominus}$) easily attack so the reaction rate is faster comparatively.



Here nucleophile ($-SC_6H_5$) can not easily attack because of steric hindrance


H+
HNO2
Equatorial amine
OH⁻
Major
HNO3
H+
Axial amine
Major

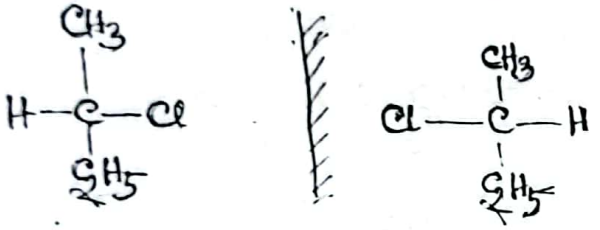
* Explain the following term:-

Symmetric molecule:

A symmetric molecule is one which passes an alternating axis of symmetry and therefore it is optically active

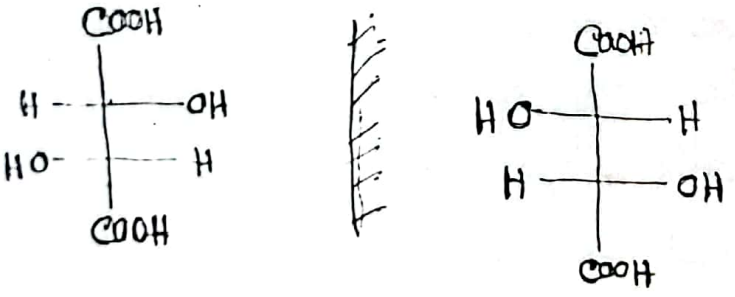
Asymmetric molecule:

A symmetric molecule is one which does not passes an alternating axis of symmetry and also lack of simple axis of symmetry and usually optically active. A chiral carbon containg molecule is always asymmetric



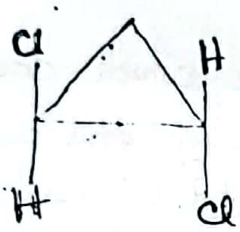
Disymmetric molecule:-

A disymmetric molecule is one which does not passes an alternating axis of symmetry but passes a simple axis of symmetry and usually optically active



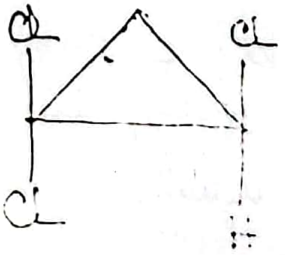
Spe fy the compound trans-1,2 dichlorocyclopropan and 1,1,2-Trichlorocyclopropane as asymmetric and disymmetric

Trans-1,2-dichlorocyclopropane :



Asymmetric molecule because of the following
(i) Absence of simple axis of symmetry
(ii) Absence of Alternating axis of symmetry

1,1,2-Trichlorocyclopropane :

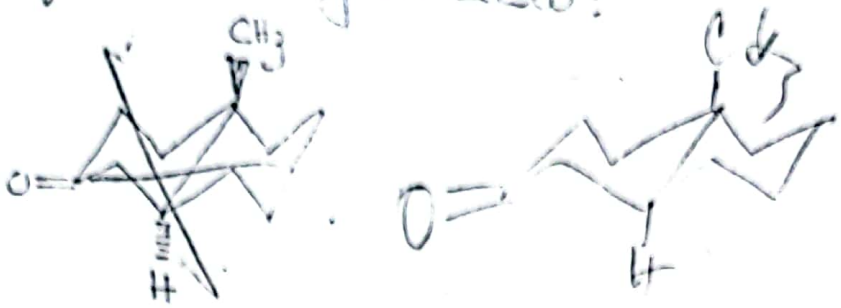


Disymmetric molecule because it contain one simple axis of symmetry but do Contain alternating axis of symmetry

Molecules that are not superimposable on the mirror image are calle disymmetric

Draw the conformation of Trans-10-methyldecan-2-one

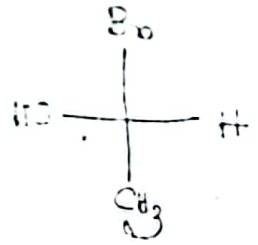
The conformation is given below:



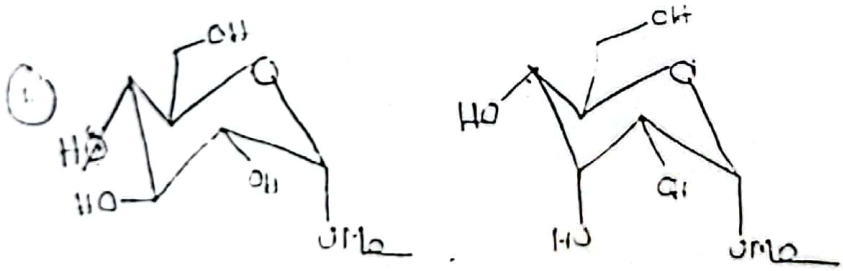
Which molecules types of molecules possess C_∞ axis.

Linear molecules posses C_∞ axis



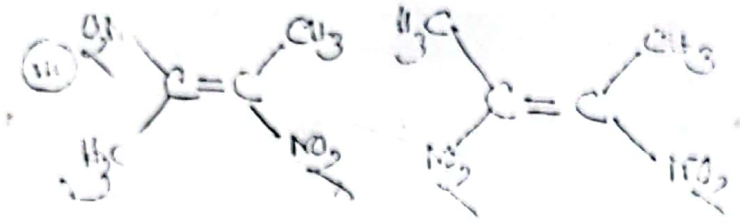


Enantiomers



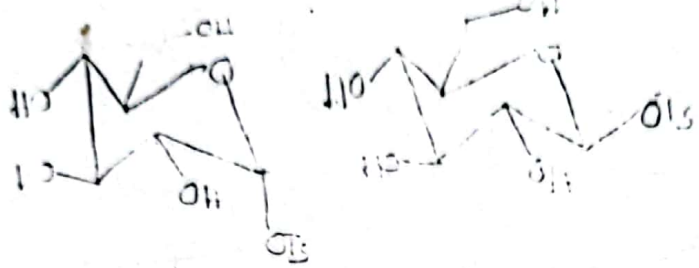
Epimers

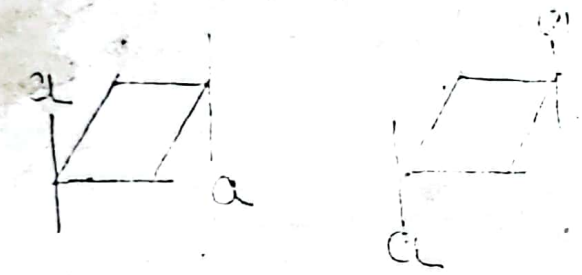
(III)



Cis - Trans

(IV)





Conformational isomers